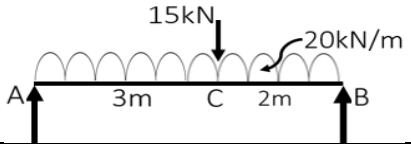
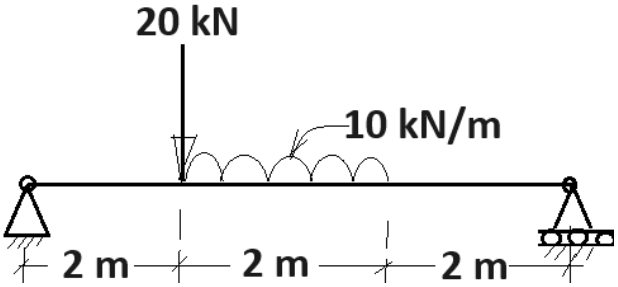
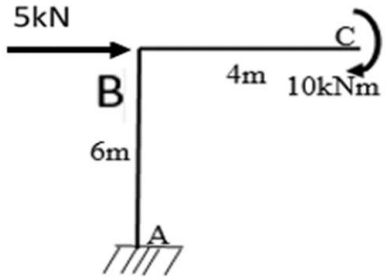
MANIPAL INSTITUTE OF TECHNOLOGY
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(A constituent unit of MAHE, Manipal)

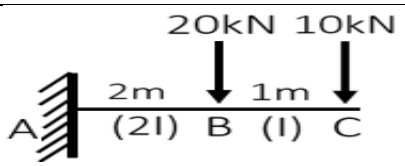
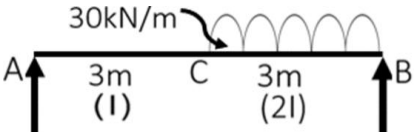
DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOV 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**CIVIL ENGINEERING**	**Course Type**	**C**
Course Name	**MECHANICS OF STRUCTURES**	**Course Code**	**2124**
Semester	**III**	**Max Marks**	**50**
Date	**28-12-2024**	**Duration**	**3 HRS**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	A simply supported beam of span 5m loaded with UDL 20kN/m on entire span and point force 15kN at3m from left support. Analyse and draw the Shear Force and Bending Moment Diagrams. Locate the points of maximum bending moment. 	**5**	1	1,2, 12	**4**
1B	A cantiliver beam of span 3m having cross-section 300 mm(wide) x 600 mm(deep). If the allowable bending stresses in the material is limited to 50 MPa, analyze maximum Point force at mid span that the beam can carry.	**3**	2	1,2, 12	**4**
1C	A circular bar of length 300 mm and diameter 12 mm is having capacity to resist shear stress of 30MPa. Analyse for torque carrying capacity of the bar.	**2**	2	1,2, 12	**3**
2A	 Using castiglianos strain energy Method analyse for Deflection at 4m from left support. I = 10 x 10^{-6} m^4 and E = 200 GPa	**5**	4	1,2, 12	**4**
2B	At a point in a strained material there is a tensile stress of 60 MPa upon the horizontal planes and compressive stress of 30MPa upon the vertical planes. Shear stress of 20 MPa is	**3**	2	1,2, 12	**4**

	acting on all these planes. Determine the planes of principal stress and magnitude of the principal stress.				
2C	A solid shaft is to transmit 250kW at 150 rpm. If the maximum angle of twist is limited to 2^0 per meter length, calculate the diameter of the shaft.G=80GPa.	**2**	2	1,2, 12	**3**
3A	Using castiglianos method, analyse horizontal displacement of the point C of vertical bent, ABC shown in the figure. Moment of inertia for the member BC is I and that for AB 2I. 	**5**	5	1,2, 12	**4**
3B	The cross section of a cantilever beam of span 3m is a T-section (kept upright with flange on the top) of overall depth 400mm and flange width 200mm, thickness of both flange and rib is 20mm.If the extreme fibre stress in bending compression is limited to 45 MPa, calculate the maximum UDL the beam can carry inclusive of self-weight. I_{NA} = 196.41 X 10^6 mm^4, NA is at 258.96 mm from bottom most fibre.	**3**	2	1,2, 12	**4**
3C	With example explain section modulus of a flexural member.	**2**	2	1,2, 12	**2**
4A	Compare the Euler's load & Rankine's buckling load for a hollow cylindrical cast iron column, 8 m long, 180 mm external diameter and 25 mm thick, and fixed at both ends. E = 80 GPa. Use σ_c = 550 MPa and a = 1/600.	**4**	3	1,2, 12	**4**
4B	Determine the deflection under the point load on a simple supported beam of span 6m loaded with a clockwise moment 20kN-m at 3m from left support. Take EI = 9000kN-m2. Use Macaulay's double integration method.	**3**	4	1,2, 12	**4**
4C	Determine maximum deflection at free end of cantilever beam 3m in length subjected to point force 25 KN at mid span using unit load method. EI = 12000kN-m^2	**3**	5	1,2, 12	**4**
5A	Using moment area method analyse for deflection and slope at free end of cantilever beam. assume I = 8 x 10^{-6} m^4 and E = 200 GPa	**4**	4	1,2, 12	**4**

					
5B	Analyse for vertical deflection at mid span of simply supported beam 5m length using unit load method subjected to UDL 30kN/m on right half of the span. 	**4**	5	1,2, 12	**4**
5C	Define Maxwell's reciprocal theorem and Bettis Law used in unit load method.	**2**	5	1,2, 12	**2**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***5***
2	**15**
3	***4***
4	***12***
5	***14***

BL	Total Marks
2	**4**
3	**4**
4	**42**
5	